

1. PROJECT & TEAM DETAILS

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2. PROBLEM DOMAIN IDENTIFICATION & ANALYSIS

Weightage: 3 Marks

Technical Focus Area: Agentic AI & Advanced Deep Learning (DL) | Explainable AI (XAI)

Problem Statement: Traditional blood banking and hospital supply chains operate as disconnected silos, relying on manual phone inquiries and static spreadsheets, causing critical bottlenecks:

- Inventory Asymmetry & Localized Stockouts:** Hospitals face acute shortages of rare blood groups (e.g., O-, B-) during surgeries while neighboring centers retain surplus units with zero real-time visibility.
- Perishability & High Wastage Paradox:** Blood components have strict shelf lifespans (platelets 5 days, red cells 42 days). Over-buffering causes substantial shelf expiry (~10–15%) alongside simultaneous shortages.
- Suboptimal & Reactive Logistics:** Inter-facility blood transfers occur ad-hoc during emergencies without routing optimization, travel-time factoring, or shelf-life-aware stock prioritization.
- Delayed Emergency Donor Mobilization:** Mass-broadcast donor outreach lacks precision geospatial proximity filtering, antigen compatibility ranking, and availability probability modeling.

Solution Vision & Technical Focus Areas: *LifeLink AI* bridges this gap with an autonomous citywide Command Center simulating 30 hospitals and 10 blood banks, incorporating four core AI paradigms: • **Agentic AI:** Multi-agent autonomous rebalancing & emergency donor recruitment dispatch. • **Advanced Deep Learning (DL):** PyTorch Temporal Fusion Transformers (TFT) & Bi-LSTM for 24/48/72h multi-horizon demand forecasting. • **Explainable AI (XAI) & GenAI:** Gemini 2.0 Flash / OpenAI LLM layer translating MILP optimization math into human-readable clinical justifications.

3. PROJECT SCOPE & SYSTEM BOUNDARIES

- In-Scope Features:** Multi-hospital synthetic network simulation (30 hospitals, 10 blood banks, 8 blood groups), 24h/48h/72h DL demand forecasting, dynamic shortage risk computation, Google OR-Tools MILP transfer optimization, Agentic donor dispatch, interactive GIS command center UI, and XAI clinical justifications.
- Out-of-Scope (Boundaries):** Direct write-back integrations into production hospital EMR/HIS databases without clinician authorization, physical cold-chain hardware telematics, and automated clinical serological cross-matching.

4. ARCHITECTURE & TECHNOLOGY STACK

Frontend & UI	Next.js 15 (App Router), React 19, TypeScript, TailwindCSS v4, Leaflet / Mapbox GL, Framer Motion
Optimization Engine	Google OR-Tools (Mixed Integer Linear Programming - MILP), NetworkX, Route ETA Matrix
AI Focus Areas (DL, XAI, GenAI, Agentic)	PyTorch (TFT & Bi-LSTM DL), Agentic Dispatch Logic, XGBoost/LightGBM, XAI/GenAI (Gemini 2.0 Flash / OpenAI API)
Database & Simulation	PostgreSQL, PostGIS (Geospatial queries), SQLAlchemy / Drizzle ORM, Synthetic City Generator (Python)

5. STEPS TO EVOLVE (PROJECT ROADMAP)

PHASE 1 (M1) Domain & Core Setup Problem domain analysis, synthetic data generator (30 hospitals, 10 banks), PostgreSQL schema.	PHASE 2 (M2) Demand & Shortage Engine LSTM & XGBoost 24/48/72h demand models, baseline comparison (MA), shelf-life shortage alert layer.	PHASE 3 (M3) Optimization & Donors Google OR-Tools MILP multi-node transfer rebalancing, donor matching, LLM clinical justification.	PHASE 4 (M4) Command UI & Testing Next.js GIS Command Center, stress-testing (demand spikes, disasters), benchmark vs traditional.
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